

Intercepts

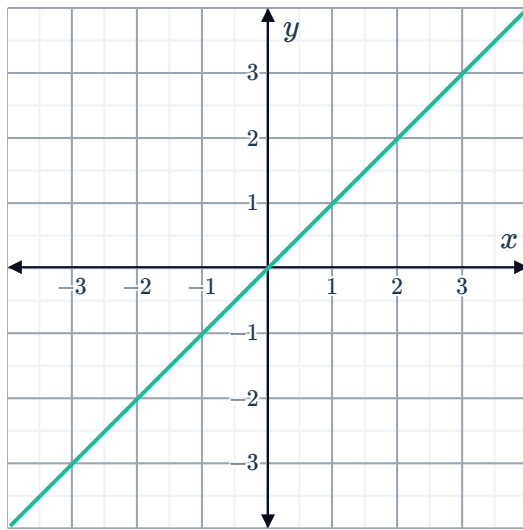


1 For each of the following graphs:

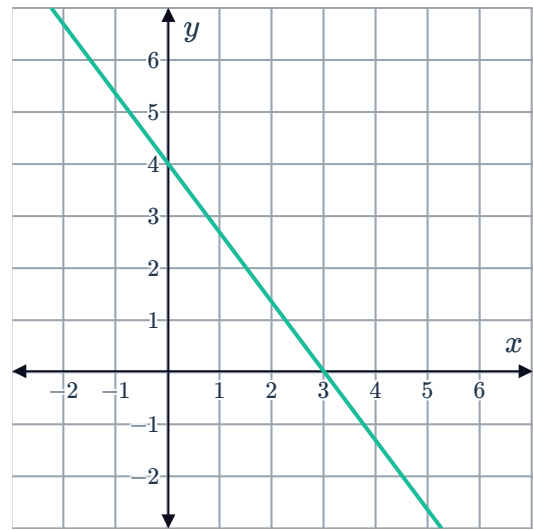
i State the value of the x -intercept.

ii State the value of the y -intercept.

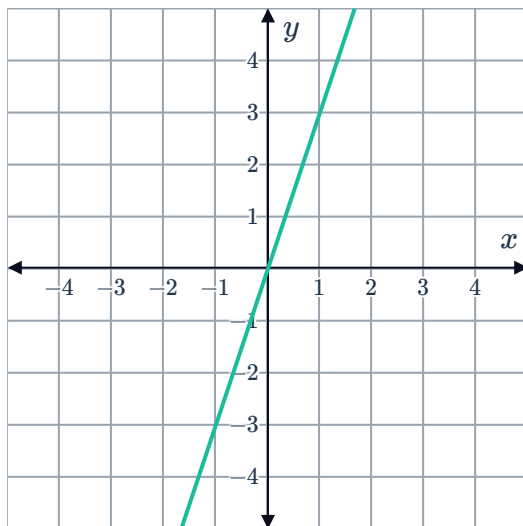
a



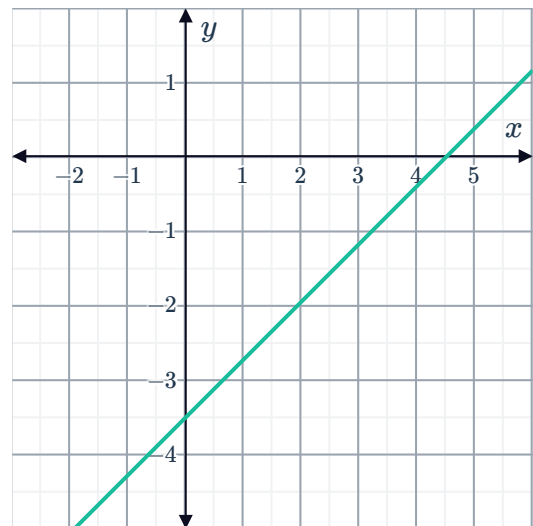
b



c

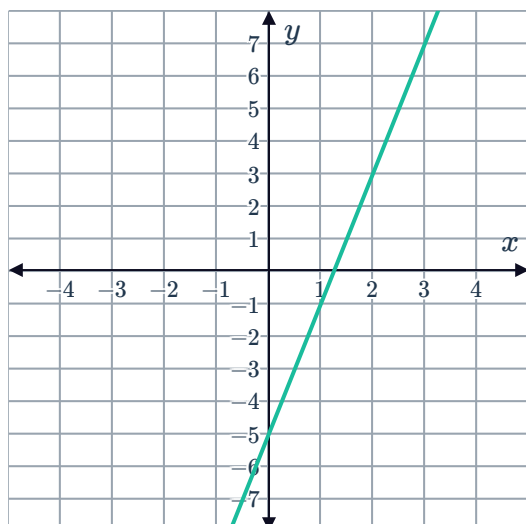


d

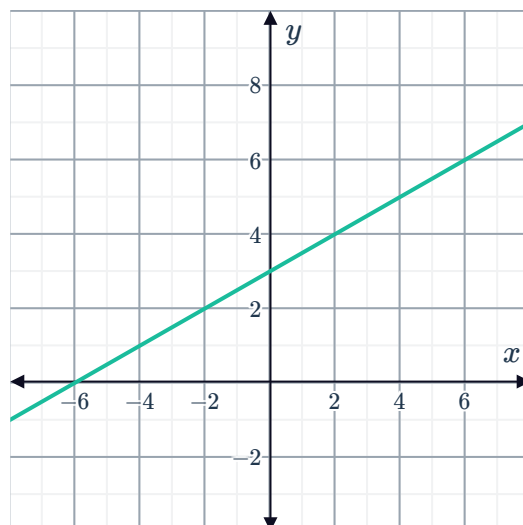


2 Given each linear equation and its graph, find the coordinates of the y -intercept:

a $y = 4x - 5$

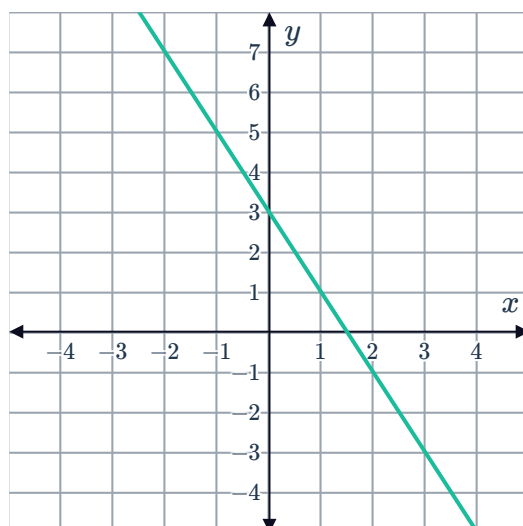


b $y = \frac{x}{2} + 3$



3 Consider the following graph of the line $y = -2x + 3$:

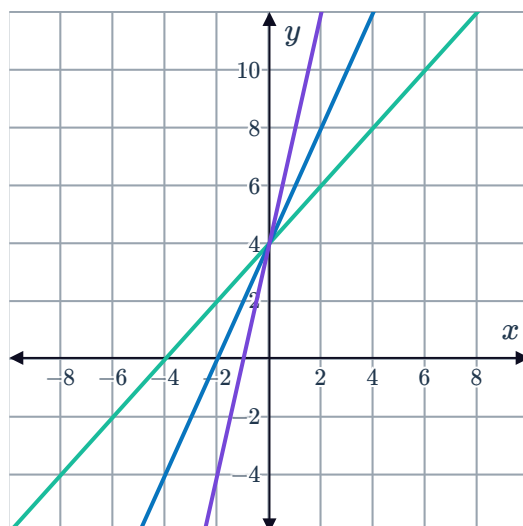
- State the y -value, when x is 0.
- Explain the relationship between the value of the y -intercept and the equation of the line.
- If the equation of a line is $y = mx + c$, state the value of the y -intercept.



4 Consider the following three linear equations and their corresponding graphs:

$$y = x + 4, y = 2x + 4, y = 4x + 4$$

- What do all of the equations have in common?
- What do all of the graphs have in common?
- What conclusion can be made about lines that have the form $y = mx + 4$?



5 Find the value of the y -intercept of the following lines:



a $y = 7x + 3$

b $y = 3x - 5$

c $y = -8x - 3$

d $y = -8x + 4$

e $y = 9x$

f $y = 2$

g $y = 2x + \frac{2}{3}$

h $y = \frac{3x + 8}{5}$

6 Determine whether the following equations represent lines that will cross the y -axis at 2:

a $y = 5x + 2$

b $y = 2 - x$

c $y = 2x$

d $y = 2x - 4$

e $y = x + 2$

f $y = x - 2$

g $y = 2$

h $y = \frac{x + 4}{2}$

7 The x -intercept occurs when $y = 0$. Find the value of the x -intercept for the following lines:

a $y = 2x - 2$

b $3x + y = -3$

c $y = 4x - 8$

d $2y + x = -3$

e $y = 9x$

f $2y + 2x = 4$

g $3x - 5y = 1$

h $x = \frac{3y + 10}{5}$

8 For each of the following equations:

ii Find the coordinates of the y -intercept.

i Find the coordinates of the x -intercept.

iii Use the intercepts to sketch the graph of the line.

a $y = 2x - 4$

b $y = -2x + 2$

c $y = 3x - 3$

d $y = -4x + 8$

Lines through the origin

9 Consider the linear equation $y = 5x$.

a Find the coordinates of the y -intercept.

b Find the coordinates of the x -intercept.

c Find the value of y when $x = 2$.

d Hence, sketch the graph of the line.

10 Consider the linear equation $y = -\frac{5x}{4}$.

a Find the coordinates of the y -intercept.

b Find the coordinates of the point on the line where $x = 4$.

c Hence, sketch the graph of the line.

11 If a line has equation $y = mx + c$, explain how you can tell if the line will pass through the



12 Determine whether the following equations represent lines that will pass through the origin:

a $y = 8x - 8$

b $y = \frac{x}{8}$

c $y = -6x$

d $y = 8x$

e $y = \frac{x}{8}$

f $y = 0$

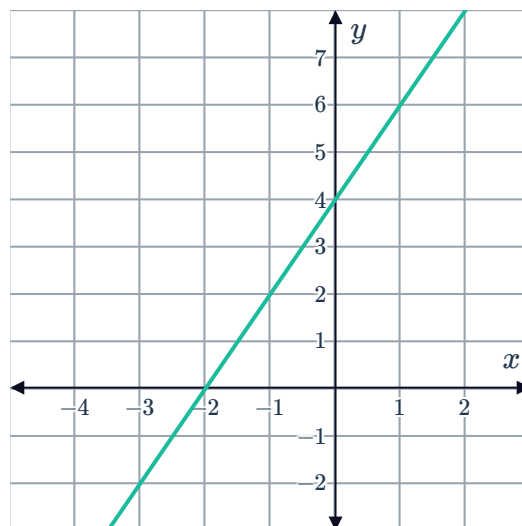
g $y = -x$

h $y = -6x - 6$

Gradient

13 Consider the line graph shown:

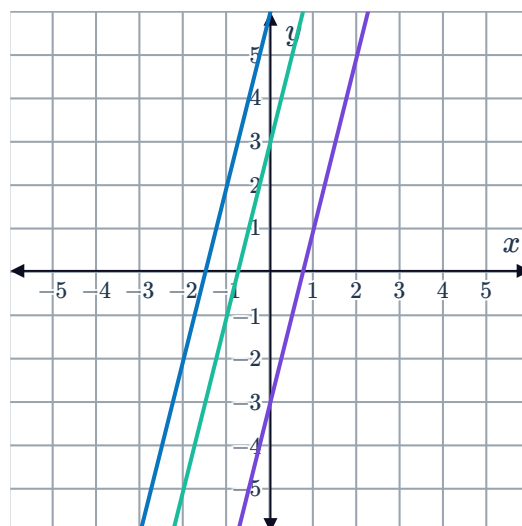
- a State the y -value when $x = 0$.
- b State the y -value when $x = 1$.
- c When the x -value increases by 1, by how much does the y -value change?
- d Hence state the gradient of the line, m .
- e The equation of this line is $y = 2x + 4$. Explain how to find the gradient from the equation of the line.



14 Consider the following three linear equations and their corresponding graphs:

$$y = 4x + 3, y = 4x + 6, y = 4x - 3$$

- a What do all of the equations have in common?
- b What do all of the graphs have in common?
- c What conclusion can be made about lines that have the form $y = 4x + c$?



15 Find the gradient, m , of the following linear equations:

a $y = 9x + 3$

b $y = -7x + 5$

c $y = \frac{5x}{4} + 2$

d $y = -x + 5$

16 From the following list of equations, select lines that have the same gradient:



- $y = 2 + 7x$
- $y = \frac{x}{7} + 2$
- $y = 5 - 7x$
- $y = 7x$
- $y = 5x + 7$
- $y = 7x - 2$

17 For each linear equation:

i Find the value, m , of the gradient.

ii Find the value, c , of the y -intercept.

a $y = 2x + 9$

b $y = 5x - 6$

c $y = -5x + 8$

d $y = -4x - 2$

e $2y = 8x - 1$

f $3y = -6x - 2$

g $y - 5x = 4$

h $2y - 3x = 6$

18 Given the values of m and c , write the equation of the line:

a $m = 2, c = 5$

b $m = -3, c = 2$

c $m = -2, c = -1$

d $m = 4, c = 0$

e $m = 0, c = -7$

f $m = 0, c = 4$

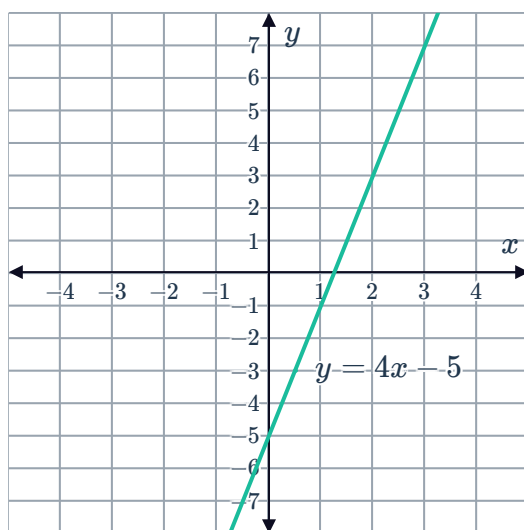
g $m = \frac{1}{2}, c = -2$

h $m = -\frac{3}{4}, c = \frac{1}{2}$

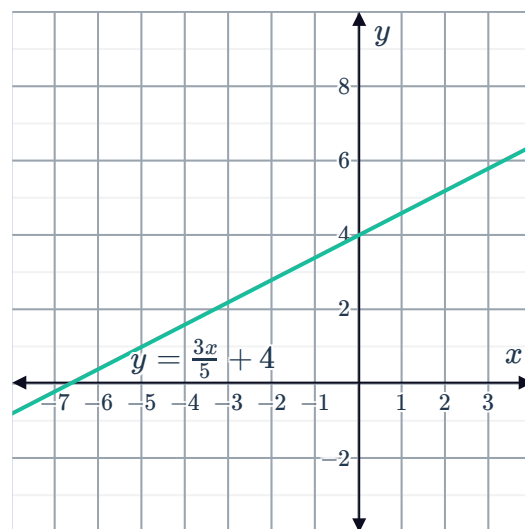
Approximate solutions of equations

19 Approximate the solutions of the following equations using the graphs provided. Round your answers to one decimal place.

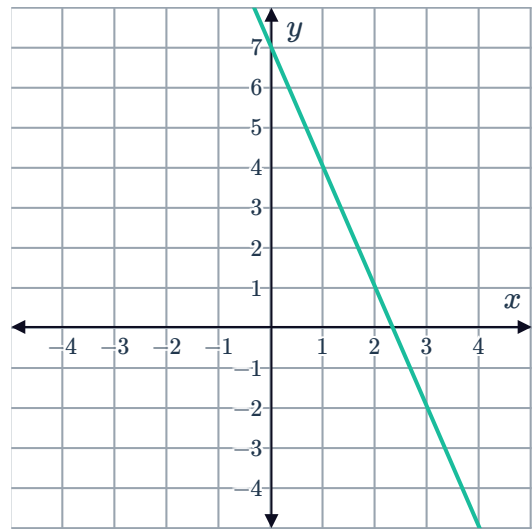
a $4x - 5 = 0$



b $\frac{3x}{5} + 4 = 0$



-  **20** Consider the following graph of the line $y = -3x + 7$:



- a** Estimate the x -intercept to one decimal place.
 - b** Hence, estimate the solution to the equation $-3x + 7 = 0$ to one decimal place.
 - c** Estimate the x -coordinate of the point on the line where $y = 6$.
 - d** Hence, estimate the solution to the equation $-3x + 7 = 6$ to one decimal place.
- 21** Estimate the solutions of the following equations by first drawing the appropriate graph. Round your answers to one decimal place.

a $5x - 11 = 0$

b $-4x + 9 = 0$

c $6x - 9 = 0$

d $\frac{9x}{5} - 1 = 0$

e $-\frac{7x}{2} + 10 = 0$

f $\frac{3x}{8} - 5 = 0$

g $4x + \frac{1}{2} = 0$

h $\frac{2x}{3} + 7 = 0$